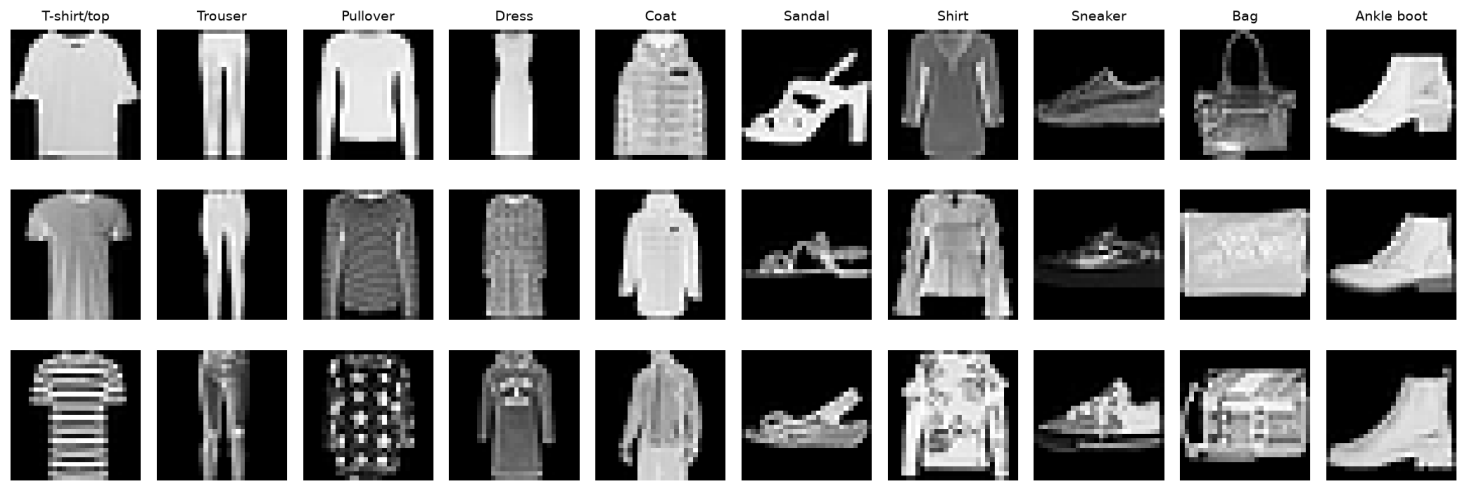
Inventory Image Tagging

One clothing image, one category

Phase 1: compare simple classifiers on Fashion-MNIST



28 × 28 grayscale images. Ten categories, including shirts, bags and shoes.

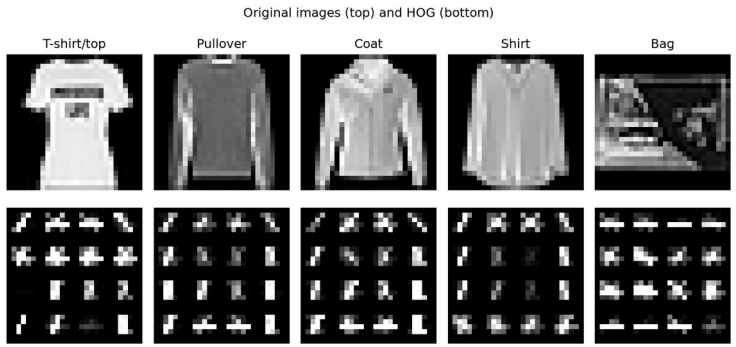
Aryan, Divyanshu Singh, Jatin Kumar Singh, Jatin Verma, Kushagra Bhardwaj

Data and methods

Data split	Images
Train models	48,000
Choose settings	12,000
Final test	10,000

Pixels: 784 brightness values

HOG: 324 values describing edge directions



Same data for all eight combinations

Two ways to describe images × four classifiers = eight models

Logistic Regression, k-NN, Random Forest and Linear SVM

Results on the final test set

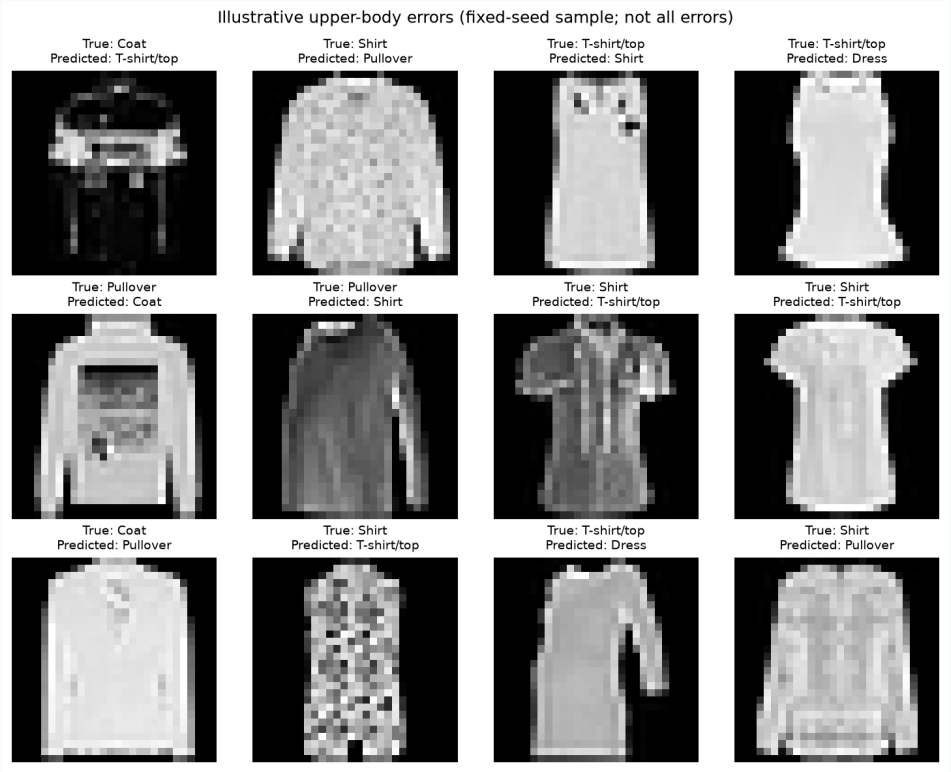
Classifier	Pixels accuracy	HOG accuracy	Pixels macro F1	HOG macro F1
Random Forest	87.61%	84.59%	0.8747	0.8455
k-NN	85.09%	83.81%	0.8515	0.8382
Logistic Regression	83.96%	84.55%	0.8390	0.8450
Linear SVM	83.97%	84.01%	0.8373	0.8385

Selected using validation: Pixels + Random Forest

87.61% correct on test images. Macro F1: 0.8747

Median prediction: 24.98 ms. Saved model: 69.96 MiB.

Mistakes and limits



Shirt is the hardest category

592 of 1,000 shirts identified correctly

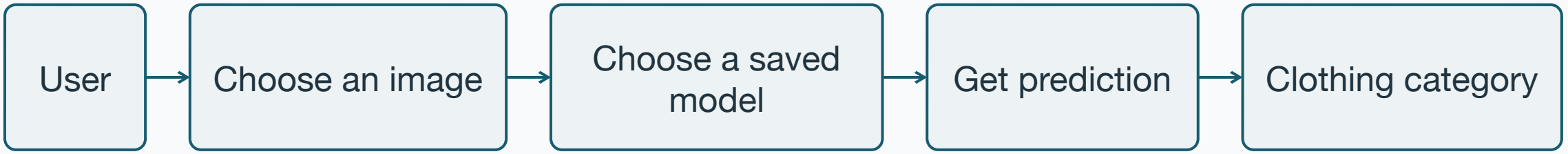
149 shirts labelled T-shirt/top
124 shirts labelled Pullover

Similar clothing outlines can lead to mistakes.

Ordinary photos have no measured accuracy. CNN work belongs to Phase 2.

Dashboard prediction flow

Start with an included sample or upload a clothing picture.



Default model: Pixels + Random Forest. Change it in settings.

Preview the prepared 28 × 28 image.
Then view the category and prediction time.

inventory-image-tagging.streamlit.app